



Gowin EDID PROM IP

User Guide

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Revision History

Date	Version	Description
10/30/2020	1.0E	Initial version published.
03/31/2023	2.0E	● Memory read/write interface added. ● Interface screenshots updated.

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1 About This Guide

1.1 Purpose

The purpose of Gowin EDID PROM IP User Guide is to help you learn this IP by providing the descriptions of features, functions and reference design, etc.

1.2 Related Documents

The latest user guides are available on GOWINSEMI Website. You can find the related document at www.gowinsemi.com.

- [DS100, GW1N series of FPGA Products Data Sheet](#)
- [DS117, GW1NR series of FPGA Products Data Sheet](#)
- [DS821, GW1NS series of FPGA Products Data Sheet](#)
- [DS861, GW1NSR series of FPGA Products Data Sheet](#)
- [DS881, GW1NSER series of FPGA Products Data Sheet](#)
- [DS891, GW1NRF series FPGA Products Data Sheet](#)
- [DS102, GW2A series of FPGA Products Data Sheet](#)
- [DS226, GW2AR series of FPGA Products Data Sheet](#)
- [DS961, GW2ANR series of FPGA Products Data Sheet](#)
- [DS976, GW2A-55 Data Sheet](#)
- [SUG100, Gowin Software User Guide](#)

1.3 Terminology and Abbreviations

Table 1-1 shows the abbreviations and terminology used in this manual.

Table 1-1 Terminology and Abbreviations

Terminology and Abbreviations	Meaning
DDC	Display Data Channel
DDWG	Digital Display Working Group
DVI	Digital Visual Interface
EDID	Extended Display Identification Data
FPGA	Field Programmable Gate Array
PROM	Programmable Read-only Memory
VESA	Video Electronics Standards Association

1.4 Support and Feedback

Gowin Semiconductor provides customers with comprehensive technical support. If you have any questions, comments, or suggestions, please feel free to contact us directly by the following ways.

Website: www.gowinsemi.com

E-mail: support@gowinsemi.com

2 Overview

2.1 Overview

The Extended Display Identification Data specification is established by VESA for the display identification data. EDID is stored in the DDC memory of the display. After the computer host is connected to the display, the computer host reads EDID in the DDC memory through DDC channel.

Gowin EDID PROM IP provides a configurable ROM for you to store EDID data and an I²C interface to read the EDID data.

Table 2-1 Gowin EDID PROM IP

Gowin EDID PROM IP	
Logic Resource	Please refer to Table 2-2.
Delivered Doc.	
Design Files	Verilog (encrypted)
Reference Design	Verilog
TestBench	Verilog
Test and Design Flow	
Synthesis Software	GowinSynthesis
Application Software	Gowin Software (V1.9.7Beta and above)

Note!

For the devices supported, you can click [here](#) to get the information.

2.2 Features

- Supports I²C bus interface
- Supports memory size of 256 bytes
- Supports ROM and RAM

2.3 Resource Utilization

Gowin EDID PROM IP can be implemented by Verilog. Its performance and resource utilization may vary when the design is employed in different devices, or at different densities, speeds, or grades.

Taking GW1N-1 series of FPGA as an instance, the resource utilization is as shown in Table 2-2.

Table 2-2 Resource Utilization

Device Series	Speed Grade	Name	Resource Utilization	Note
GW1N-1	-6	LUT	168	Memory ROM mode
		REG	73	
		BSRAM	1	

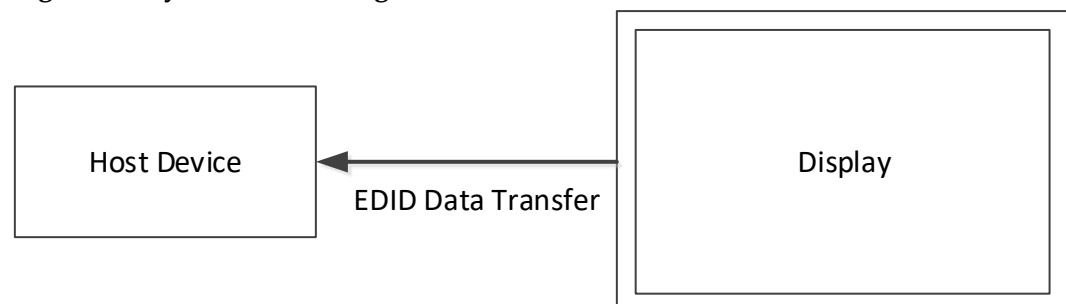
3 Functional Description

3.1 System Block Diagram

The EDID contains the display and its performance parameters, such as vendor, image size, color settings, manufacturer presets, frequency limits, display name and serial numbers.

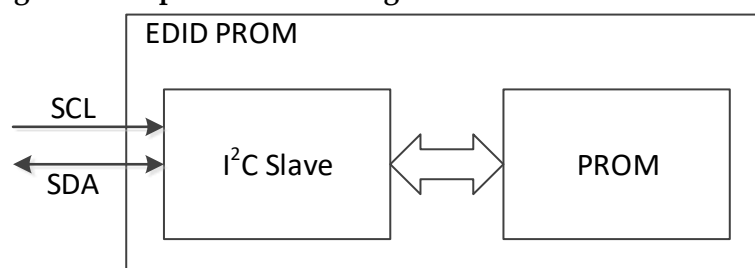
Host Device gets some display properties by reading EDID data from display. In short, EDID is a nameplate for display, and EDID data is usually transmitted through DDC channel.

Figure 3-1 System Block Diagram



3.2 Implementation Diagram

Figure 3-2 Implementation Diagram



As shown in Figure 3-2, EDID PROM IP includes I²C Slave module and PROM module.

- I²C Slave Module

It implements slave device I²C interface function and converts addresses, commands and data into PROM read/write addresses,

commands and data.

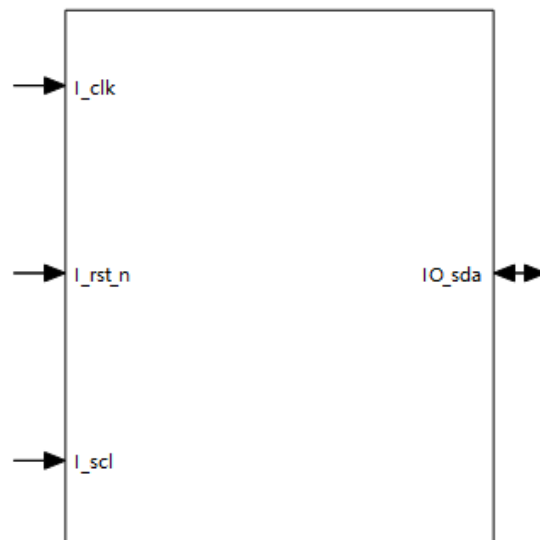
- PROM Module

It is used to store EDID data with 256 bytes.

3.3 Port List

The I/O ports of Gowin EDID PROM IP are shown in Figure 3-3.

Figure 3-3 Gowin EDID PROM IP I/O Diagram



Ports vary slightly depending on the parameters. The details of I/O ports of Gowin EDID PROM IP are shown in Table 3-1.

Table 3-1 I/O List

No.	Signal Name	I/O	Description	Note
1	I_clk	I	System clock, 5~200MHz.	The I/O of all the signals takes EDID PROM IP as reference.
2	I_rst_n	I	Reset signal, active-low.	
3	I_scl	I	I ² C clock signal for DDC channel	
4	IO_sda	IO	I ² C data signal for DDC channel	
5	I_mema_clk	I	Memory read/write clock, valid in RAM mode.	
6	I_mema_rd	I	Memory read enable, valid in RAM mode.	
7	I_mema_we	I	Memory write enable, valid in RAM mode.	
8	I_mema_addr	I	Memory address, valid in RAM mode.	
9	I_mema_di	I	Memory write data, valid in RAM mode.	
10	O_mema_do	O	Memory read data, valid in RAM mode.	

3.4 Parameters Configuration

Table 3-2 Gowin EDID PROM IP Parameters

No.	Name	Range	Default	Description
1	Memory Mode	ROM/RAM	ROM	Memory mode: ROM/RAM
2	Memory Initialization File	-	-	Memory Initialization File

3.5 Introduction to EDID

One EDID data is composed of 128-byte base block (Block0) and 128-byte extended block (Block 1). Block0 is required and Block1 is optional. Each byte represents information about display. For detailed descriptions, see [VESA ENHANCED EXTENDED DISPLAY IDENTIFICATION DATA STANDARD](#).

The description of Block0 is as shown in Figure 3-4.

Figure 3-4 Block0 Structure Description

Address	Bytes	Description	Format
00h	8	Header: = (00 FF FF FF FF FF FF 00)h	See Section 3.3
08h	10	Vendor & Product Identification:	See Section 3.4
08h	2	ID Manufacturer Name	ISA 3-character ID Code
0Ah	2	ID Product Code	Vendor assigned code
0Ch	4	ID Serial Number	32-bit serial number
10h	1	Week of Manufacture	Week number or Model Year Flag
11h	1	Year of Manufacture or Model Year	Manufacture Year or Model Year
12h	2	EDID Structure Version & Revision:	See Section 3.5
12h	1	Version Number: = 01h	Binary
13h	1	Revision Number: = 04h	Binary
14h	5	Basic Display Parameters & Features:	See Section 3.6
14h	1	Video Input Definition	See Section 3.6.1
15h	1	Horizontal Screen Size or Aspect Ratio	Listed in cm. → Aspect Ratio --- Landscape
16h	1	Vertical Screen Size or Aspect Ratio	Listed in cm. → Aspect Ratio --- Portrait
17h	1	Display Transfer Characteristic (Gamma)	Binary --- Factory Default Value
18h	1	Feature Support	See Section 3.6.4
19h	10	Color Characteristics:	See Section 3.7
19h	1	Red/Green: Low Order Bits	Rx1 Rx0 Ry1 Ry0 Gx1 Gx0 Gy1 Gy0
1Ah	1	Blue/White: Low Order Bits	Bx1 Bx0 By1 By0 Wx1 Wx0 Wy1 Wy0
1Bh	1	Red-x: High Order Bits	Red-x Bits 9 → 2
1Ch	1	Red-y: High Order Bits	Red-y Bits 9 → 2
1Dh	1	Green-x: High Order Bits	Green-x Bits 9 → 2
1Eh	1	Green-y: High Order Bits	Green-y Bits 9 → 2
1Fh	1	Blue-x: High Order Bits	Blue-x Bits 9 → 2
20h	1	Blue-y: High Order Bits	Blue-y Bits 9 → 2
21h	1	White-x: High Order Bits	White-x Bits 9 → 2
22h	1	White-y: High Order Bits	White-y Bits 9 → 2
23h	3	Established Timings	See Section 3.8
23h	1	Established Timings I	
24h	1	Established Timings II	
25h	1	Manufacturer's Reserved Timings	
26h	16	Standard Timings: Identification 1 → 8	See Section 3.9
36h	72	18 Byte Data Blocks	See Section 3.10
36h	18	Preferred Timing Mode	
48h	18	Detailed Timing # 2 or Display Descriptor	
5Ah	18	Detailed Timing # 3 or Display Descriptor	
6Ch	18	Detailed Timing # 4 or Display Descriptor	
7Eh	1	Extension Block Count N If Block Maps are used then 00h ≤ N ≤ FEh and FFh is invalid. If Block Maps are not used then 00h ≤ N ≤ FFh.	Number of (optional) 128-byte EDID EXTENSION blocks to follow – if Block Maps are used then 254 is the maximum value of 'N'. If Block Maps are not used then 255 is the maximum value of 'N'.
7Fh	1	Checksum C 00h ≤ C ≤ FFh	The 1-byte sum of all 128 bytes in this EDID block shall equal zero

The description of Block1 is from the CEA-861 specification and its data structure is shown in Figure 3-5.

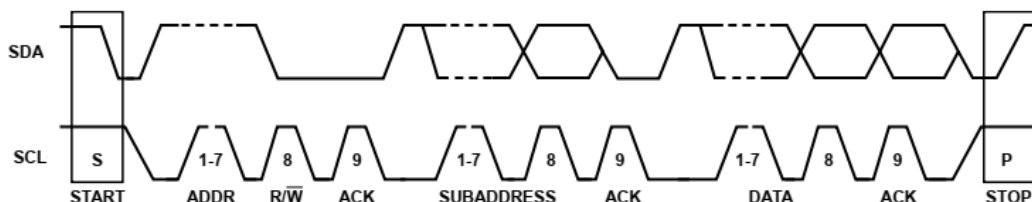
Figure 3-5 Block1 Structure Description

Byte #	Value	Description	Format
0	0x02	Tag (0x02)	
1	0x03	Revision Number	
2		Byte number offset d where 18-byte descriptors begin (typically Detailed Timing Descriptors)	d = offset for the byte following the reserved data block. If no data is provided in the reserved data block, then $d=4$. If $d=0$, then no detailed timing descriptors are provided and no data is provided in the reserved data block collection.
3		Total number of Detailed Timing Descriptors describing Native Video Formats in entire E-EDID structure. Also, indication of underscan support, audio support, and support of YCbCr is included	bit 7 (underscan) = 1 if Sink underscans IT Video Formats by default. bit 6 (audio) = 1 if Sink supports Basic Audio. bit 5 (YCbCr 4:4:4) = 1 if Sink supports YCbCr 4:4:4 in addition to RGB. bit 4 (YCbCr 4:2:2) = 1 if Sink supports YCbCr 4:2:2 in addition to RGB. lower 4 bits = total number of native DTDs (see Section 2.2 for definition of "Native Video Format").
4		Start of data block collection	This section is used for CEA Data Block Collection (see Table 42).
$d-1$		End of data block collection.	
d		Start of 18-byte detailed timing descriptors	See Section 3.10.2 of VESA E-EDID Standard [9]
$d+(18 \cdot n)-1$		End of 18-byte detailed timing descriptors where n is the number of descriptors included	
$d+(18 \cdot n)$	0x00	Beginning of Padding	
126	0x00	End of Padding	
127		Checksum	This byte should be programmed such that a one-byte checksum (add all bytes together) of the entire 128 byte block equals 0x00.

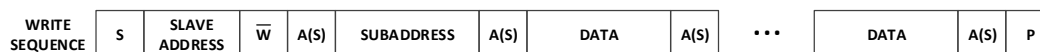
3.6 Timing Descriptions

This section describes the timing of Gowin EDID PROM IP.

The timing diagram of I²C interface is as shown in Figure 3-6.

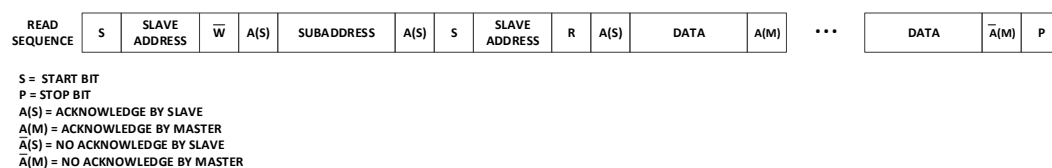
Figure 3-6 I²C Interface Timing

The write operation timing of EDID I²C is as shown in Figure 3-7.

Figure 3-7 EDID I²C Write Operation Timing

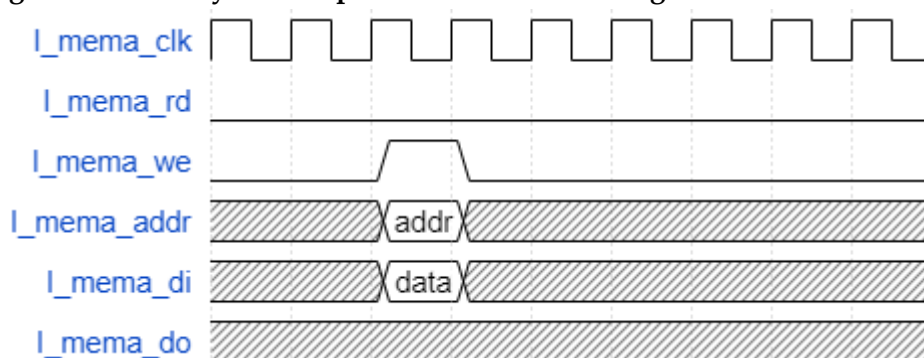
S = START BIT
 P = STOP BIT
 A(S) = ACKNOWLEDGE BY SLAVE
 A(M) = ACKNOWLEDGE BY MASTER
 $\bar{A}(S)$ = NO ACKNOWLEDGE BY SLAVE
 $\bar{A}(M)$ = NO ACKNOWLEDGE BY MASTER

The read operation timing of EDID I²C is as shown in Figure 3-8.

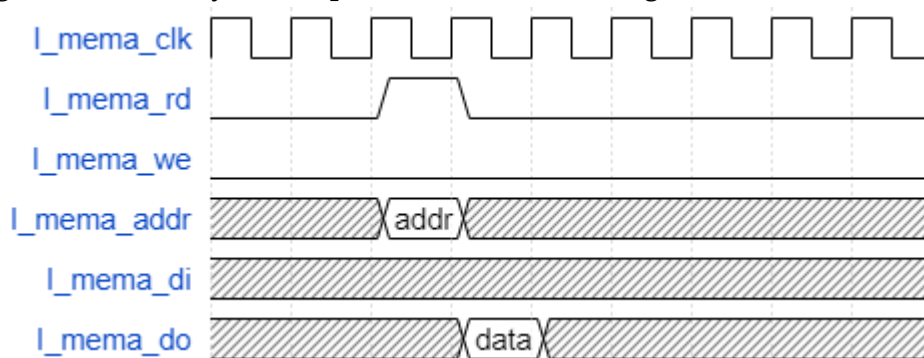
Figure 3-8 EDID I²C Read Operation Timing

Where I²C Slave Address is the 7-bit binary "1010000", corresponding to hexadecimal 0x50.

Memory write operation timing is as shown in Figure 3-9.

Figure 3-9 Memory Write Operation Interface Timing

Memory read operation timing is as shown in Figure 3-10.

Figure 3-10 Memory Read Operation Interface Timing

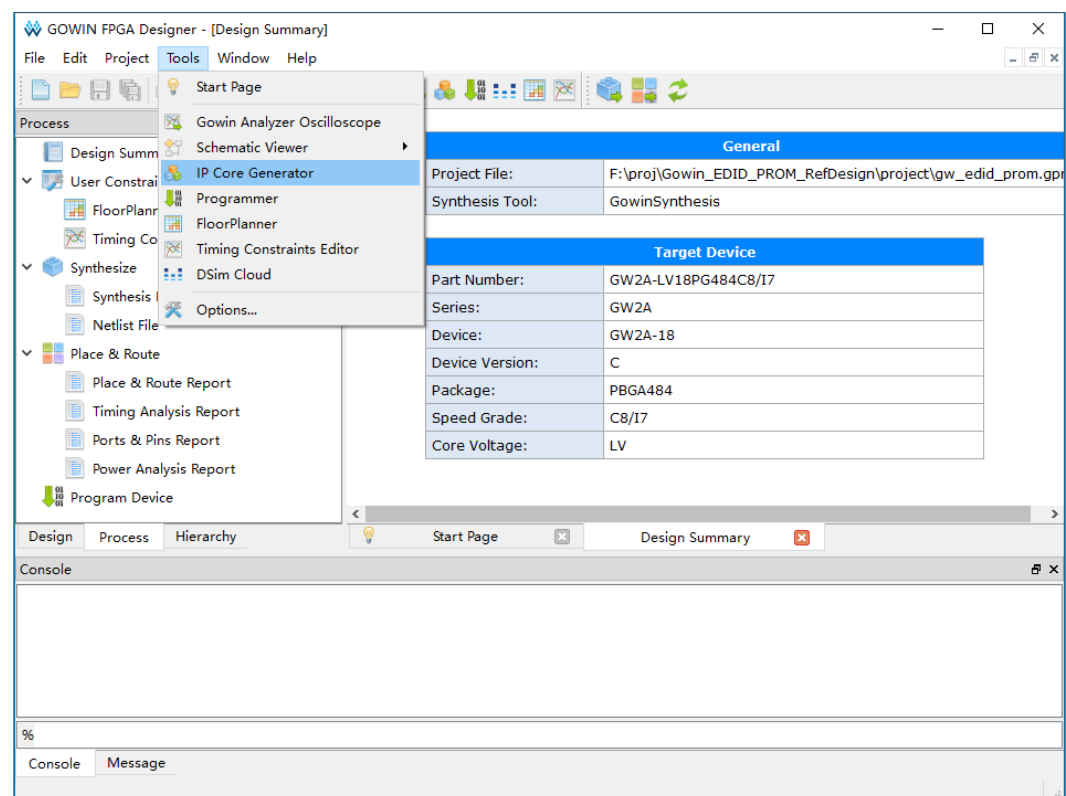
4 GUI

You can call and configure Gowin EDID PROM IP using the IP core generator tool in the IDE.

1. Open IP Core Generator

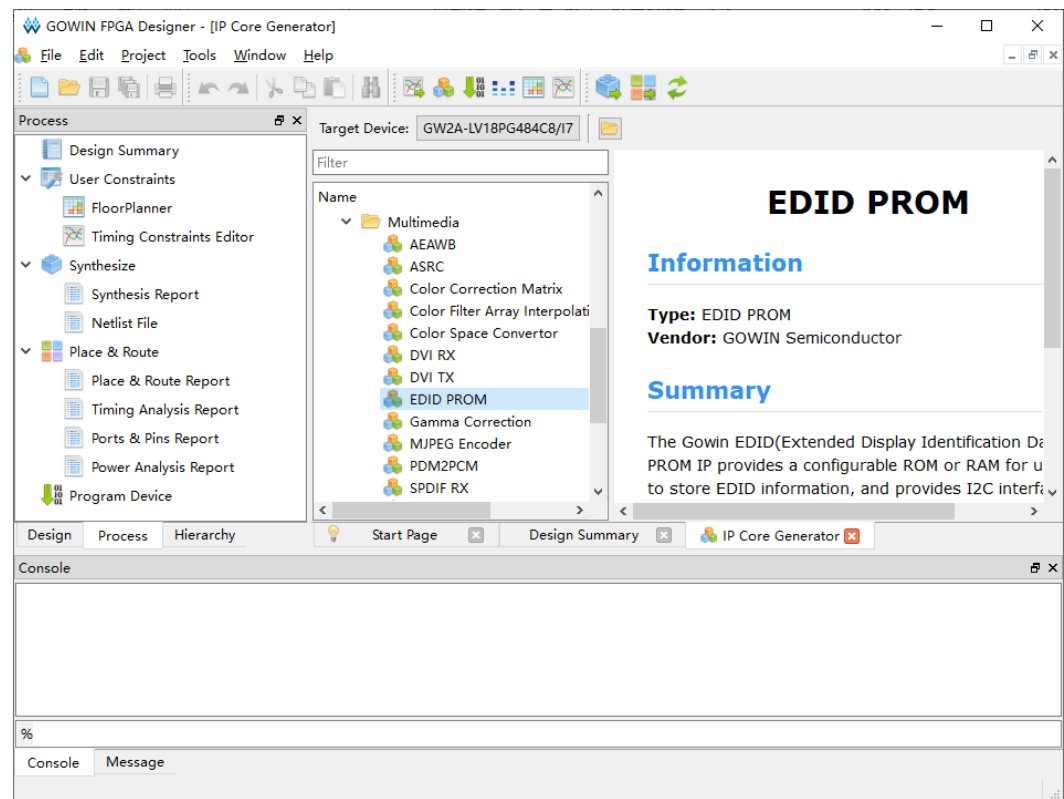
After creating the project, click the "Tools" tab in the upper left, select and open the IP Core Generator from the drop-down list, as shown in Figure 4-1.

Figure 4-1 Open IP Core Generator



2. Open EDID PROM IP Core

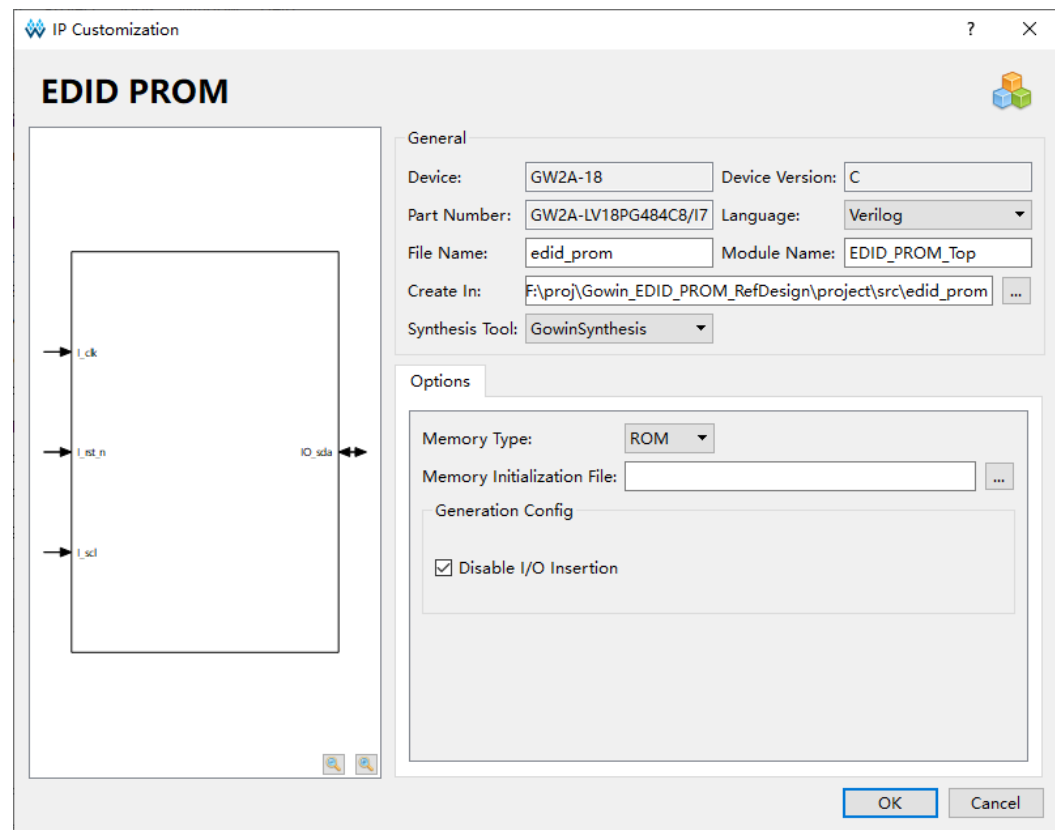
Click "Multimedia" and double click "EDID PROM" to open GUI, as shown in Figure 4-2.

Figure 4-2 Open EDID PROM IP Core

3. EDID PROM IP GUI

On the left of the GUI is the port diagram of EDID PROM IP, as shown in Figure 4-3.

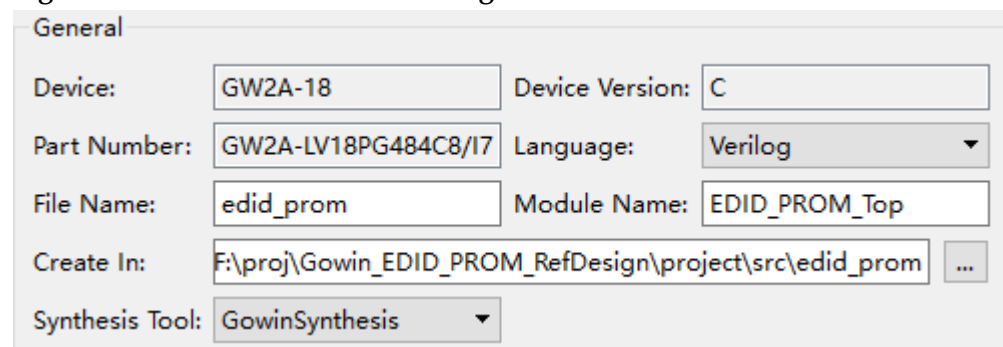
Figure 4-3 EDID PROM IP GUI



4. Parameter Configuration

See the basic information in the upper part of the GUI. Take GW2A-18C chip as an example, and select the "PBGA484" package. The "Module Name" option displays the top-level file name of the generated project, and the default is "EDID_PROM_Top". You can modify the name. The folder generated by the IP core is shown in "File Name", which contains the files required by EDID PROM IP core, and the default is "edid_prom". You can modify the path. "Creat In" displays the path of IP core folder. The default is "\project path\src\edid_prom". You can modify the path.

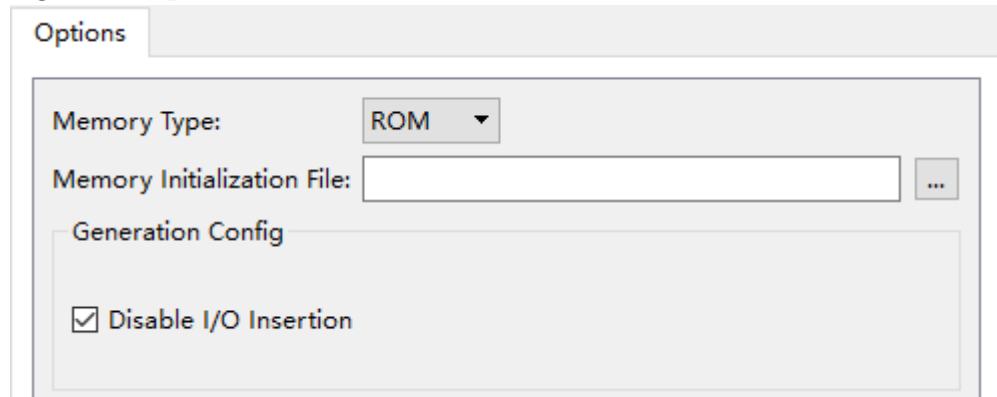
Figure 4-4 General Parameters Configuration



5. Options

You can configure memory type and memory initialization file in the Options.

Figure 4-5 Options



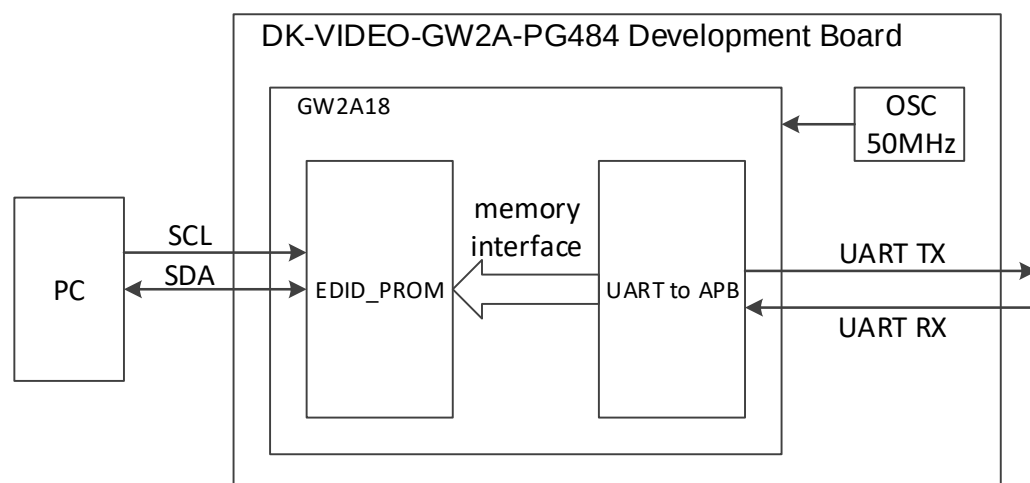
The screenshot shows a dialog box titled "Options". Inside the dialog, there is a section for "Memory Type:" with a dropdown menu currently set to "ROM". Below this is a "Memory Initialization File:" label followed by a text input field and a button with three dots "...". Underneath these fields is a section titled "Generation Config" which contains a checked checkbox labeled "Disable I/O Insertion".

5 Reference Design

This chapter focuses on the usage and structure of the reference design instance of Gowin EDID PROM IP. Please see the EDID PROM Reference Design for details at www.gowinsemi.com.

Take DK-VIDEO-GW2A-PG484 as an example, and the diagram is as shown in Figure 5-1. For details, see [DK-VIDEO-GW2A-PG484](http://www.gowinsemi.com/DK-VIDEO-GW2A-PG484) at Gowinsemi website.

Figure 5-1 Block Diagram of Reference Design



In this reference design, its steps are as follows:

1. 50MHz reference clock is taken as the sampling clock for EDID I²C.
2. Use HDMI cable to connect to the PC through the HDMI RX (J3) interface; the computer detects the connection to the display device and reads the EDID information through the DDC channel, a display device with 1280x720 recommended resolution will be recognized.
3. Use the serial debugging tool to write the new EDID information through UART, and the computer will recognize the new EDID information after unplugging the HDMI cable.

When the reference design is applied to board-level test, you can see the display device in the resolution information of the PC and also can

observe it with the on-line logic analyzer or oscilloscope.

In the simulation project provided by the reference design, simulate I²C to read EDID information, and tb is the top-level module of the simulation project.

6 File Delivery

The delivery file for Gowin EDID PROM IP includes documents, design source code and reference design.

6.1 Documents

The documents mainly contain the user guide in PDF format.

Table 6-1 Document List

Name	Description
IPUG753, Gowin EDID PROM IP User Guide	Gowin EDID PROM IP User Guide, namely this one.

6.2 Design Source Code (Encryption)

The encrypted code file contains Gowin EDID PROM IP RTL encrypted code which is used for GUI in order to cooperate with Gowin software to generate the IP core required by users.

Table 6-2 EDID PROM Design Source Code List

Name	Description
edid_prom.v	The top-level file of the IP core, which provides users with interface information, encrypted.

6.3 Reference Design

The Ref. Design folder contains the netlist file for Gowin EDID PROM IP, user reference design, constraints file, top-level file and the project file, etc.

Table 6-3 Gowin EDID PROM IP RefDesign Folder List

Name	Description
edid_test_top.v	The top module of reference design
gw_edid_prom.cst	Project physical constraints file
gw_edid_prom.sdc	Project timing constraints file
edid_800x600.mi	EDID Memory initialization file example one
edid_1024x768.mi	EDID Memory initialization file example two
edid_1280x720.mi	EDID Memory initialization file example three
edid_prom	EDID PROM IP project folder

Name	Description
edid_prom.v	Generates EDID PROM IP top file, encrypted.
edid_prom.vo	Generates EDID PROM IP netlist file
uart_apb_mst	UART to APB interface project folder
uart_apb_mst_top.v	UART to APB interface top folder
uart_apb_mst_wrapper.vo	UART to APB interface subfile, encrypted.

